Studies on apple and blueberry fruit constituents: do the polyphenols reach the colon after ingestion?

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The aim of our studies was to determine the amount of polyphenols reaching the colon after oral intake of apple juice and blueberries. After a polyphenol-free diet healthy ileostomy volunteers consumed a polyphenol-rich cloudy apple juice while others consumed anthocyanin-rich blueberries. Ileostomy effluent was collected and polyphenols were identified using HPLC-DAD as well as HPLC-ESI-MS/MS; quantification was performed with HPLC-DAD. Most of the orally administered apple polyphenols were absorbed from or metabolized in the small intestine. Between 0 and 33% of the oral dose was recovered in the ileostomy bags with a maximum of excretion after 2 h. A higher amount of the blueberry anthocyanins under study (up to 85%, depending on the sugar moiety) were determined in the ileostomy bags and therefore would reach the colon under physiological circumstances. Such structure-related availability has to be considered when polyphenols are used in model systems to study potential preventive effects in colorectal diseases.